# Computational geometry exercises

## 33-E01 Exact orientation and degeneracy

Test intersection between closed segment $a = (0, 0), b = (6, 0)$ and each of (i) $c = (2, 0), d = (8, 0)$, (ii) $c = (7, -1), d = (7, 2)$, and (iii) $c = d = (6, 0)$. Give all four orientations and the role of `on` tests. Is signed 32-bit multiplication safe for signed 32-bit coordinates?

## 33-E02 Insertions alone are insufficient

Use $a : (0, 0) \to (10, 5), b : (2, 6) \to (10, 0), c : (0, 2) \to (3, 2)$. List status and neighbor tests in endpoint order and compute the a,b intersection. Add vertical segment $v : (2, 3) \to (2, 4)$. When is its range query performed, and does it find an intersection?

## 33-E03 A detector is not an enumerator

Why does replacing the sweep's `return true` with printing and continuing invalidate the status invariant? Construct three nonvertical segments where an intersecting pair exchanges order, making the stored tree order disagree with geometric height. What new events and updates does enumeration need?

## 33-E04 Hull and boundary conventions

Use $O(0, 0), A(6, 0), B(4, 1), C(6, 4), D(2, 3), E(0, 4), F(3, 2), G(3, 0)$. Run Graham scan with only extreme vertices retained. List same-ray removals, angle order, and the orientation of each pop. Give the correct output for all-collinear input.

## 33-E05 Algebraic proof of popping

Let $O = (0, 0)$, with $r, t, u$ in strictly increasing polar-angle order over a span below $\pi$, and $D = \operatorname{orient}(O, r, u) > 0$. Prove $t = \alpha r + \beta u$, $\alpha, \beta \geq 0$, and derive $\operatorname{orient}(r, t, u) = (\alpha + \beta - 1)D$. Deduce that a nonleft turn makes t nonextreme in $CH(\{O, r, t, u\})$.

## 33-E06 Closest pairs with equal x-coordinates

Eight points have IDs 1 through 8, all x=0, and respective y-values 0,10,20,30,31,40,50,60. Split by X rank, give both recursive minimum distances, and merge. Why can partitioning by $x < \ell$ destroy the time bound? Does the seven-successor proof require strict geometric separation of the groups?

# Computational geometry exercises (continued)

## 33-E07 Arithmetic and bit costs

For n points with b-bit integer coordinates, compare arithmetic operation counts for brute-force closest pair, divide and conquer that re-sorts Y at every level, and presorting with stable partitioning. Why can squared distances preserve exact comparisons without roots? Which multiplications cease to be constant time for arbitrary b?

### References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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